VALUE ADDED PROBLEM 2:

Problem 2: Given a string s, composed of different combinations of '(' , ')', '{', '}', '[', ']'. Determine whether the Expression is balanced or not. An expression is balanced if: 1. Each opening bracket has a corresponding closing bracket of the same type. 2. Opening brackets must be closed in the correct order. Concept: Stack

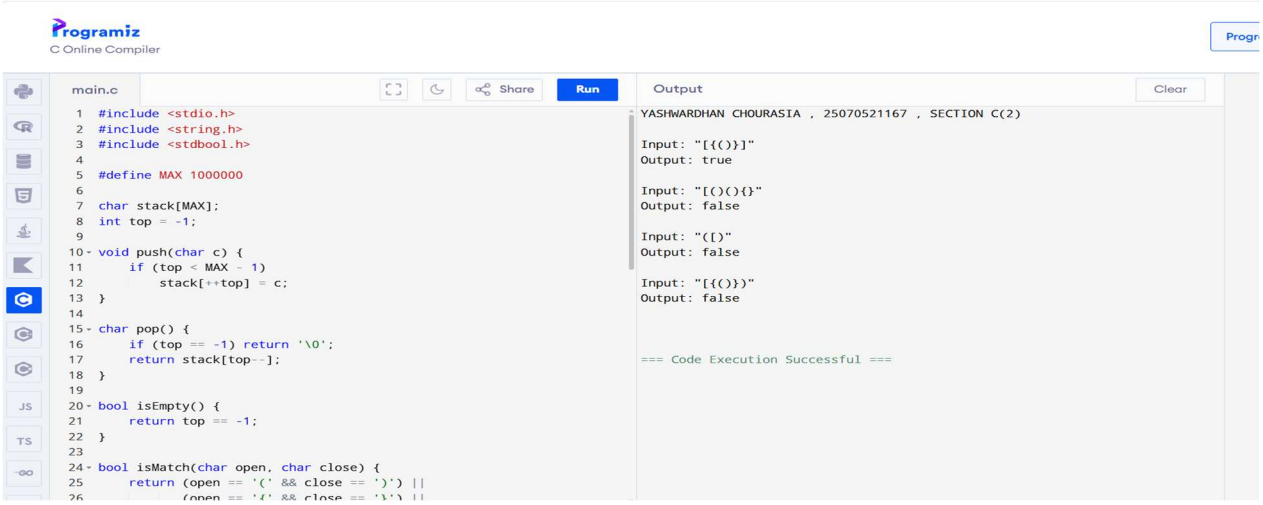
Examples :

Input: s = "[{()}]"
Output: true
Explanation: All the brackets are well-formed.

Input: s = "[()()]{}"
Output: true
Explanation: All the brackets are well-formed.

Input: s = "(]"
Output: false
Explanation: The expression is not balanced as there is a missing ')' at the end.

Input: s = "({])"
Output: false
Explanation: The expression is not balanced as there is a closing ']' before the closing '}'.

```
#include <stdio.h>

#include <string.h>

#include <stdbool.h>

#define MAX 1000000

char stack[MAX];

int top = -1;

void push(char c) {

   stack[++top] = c;}

char pop() {

   if (top == -1) return '\0';

   return stack[top--];}

bool isEmpty() {

   return top == -1;}

bool isMatch(char open, char close) {

   return (open == '(' && close == ')') ||

         (open == '{' && close == '}') ||

         (open == '[' && close == ']');
```

```
}

bool isBalanced(char* s) {

   top = -1;

for (int i = 0; s[i] != '\0'; i++) {

      char ch = s[i];

 if (ch == '(' || ch == '{' || ch == '[') {

          push(ch)   }  else if (ch == ')' || ch == '}' || ch == ']') {

          if (isEmpty()) return false;

          char top_char = pop();

          if (!isMatch(top_char, ch)) return false; }

   }

  return isEmpty();

}

int main() {

  printf("YASHWARDHAN CHOURASIA , 25070521167 , SECTION

C(2)\n\n"); char* tests[] = {"[{()}]", "[()(){}", "([)", "[{()})"};

   int n = 4;

 for (int i = 0; i < n; i++) {

      printf("Input:  \"%s\"\n", tests[i]);

      printf("Output: %s\n\n", isBalanced(tests[i]) ? "true" : "false");

  }


   return 0;

}
```